

Robust Ridge-Patterned Instance Segmentation for High-Resolution Point Cloud Measurement

Ran Jia^{ID}, Junpeng Xue^{ID}, Feifei He^{ID}, Hongyang Chen^{ID}, Kelei Wang^{ID}, Jiashi Zeng^{ID}, and Yusong Meng^{ID}

Abstract—The rapid advancement of 3-D scanning technology now enables the capture of high-resolution point clouds, establishing a new data foundation for computer-aided design (CAD) model reconstruction in reverse engineering. However, existing point cloud segmentation algorithms confront three fundamental challenges: limited instance category generalization, inadequate noise resistance, and strong scenario dependency of parameters, which significantly restrict their effectiveness in industrial-grade applications. To address these issues, we propose a robust ridge-patterned instance segmentation algorithm specifically tailored for high-resolution point clouds of mechanical components with precision geometric features. Our unsupervised method consists of four main steps: the curvature coefficient calculation using a determinantal sampling (DetS) estimator and ridge point extraction via triangular threshold, the enhancement of ridge points, the adaptive-radius region growing method for surface segmentation, and the ridge point assignment via local plane fitting. Extensive experiments on synthetic (ABC-10) and real world (OmniObject3D-8) demonstrate superior performance. Our method achieves an average mIoU of 95.52% on ABC-10 dataset, outperforming traditional methods and deep learning approaches. Notably, under 10% noise conditions, it maintains a robust mIoU of 87.11%, showcasing strong noise resilience. The algorithm's parameter stability and resolution independence make it suitable for integration with the structured-light sensors in real-world measurement systems.

Index Terms—3-D instance segmentation, geometric feature extraction, high-resolution 3-D processing, noise-resistant algorithms, point cloud segmentation.

I. INTRODUCTION

WITH the rapid development of 3-D perception technology, point cloud data have become an important form of 3-D representation of objects, showing great application potential in the fields of autonomous driving, robot navigation, and augmented reality. However, due to the disorder and unstructured nature of point cloud data, accurately dividing it into multiple simple geometric primitives remains a key challenge in the field of 3-D vision. In addition, with the continuous upgrading of 3-D scanning equipment, the resolution

of point cloud data has gradually increased from low resolution in the early days to high resolution, which makes the instance segmentation of high-precision point clouds more challenging. High-precision point cloud segmentation is a critical step in 3-D measurement. It can achieve a leap from raw geometric data to structured semantic understanding, which not only provides key support for downstream tasks, such as object grasping, scene semantic modeling [1], and surface area measurement [2], but also further promotes the machine's cognitive ability of the 3-D physical world. The current point cloud instance segmentation methods can be divided into two categories: the geometry-based methods and the learning-based methods.

A. Geometry-Based Segmentation Methods

The geometry-based methods, for instance, segmentation based on point cloud geometric attributes, can be mainly divided into three categories: RANSAC-based method, parameter space-based method, and clustering-based method [3]. A representative RANSAC-based method [4] is an efficient iterative algorithm for primitive detection, capable of processing high-resolution point clouds. While this approach and its derivatives [5], [6] demonstrate effectiveness in shape reconstruction, they are fundamentally limited to predefined geometric primitives and require many parameter tuning. Parameter space-based methods mainly include methods based on the Hough transform [7] and parameter space clustering methods. In Hough transform-based methods, Hulik et al. [8] can detect continuous planes, Rabbani and Heuvel [9] can detect cylinders in point clouds, and Limberger and Oliveira [10] use spherical accumulators to apply kernel-based Hough transform [11] for 3-D plane detection. Extending the Hough transform paradigm, the work in [12] presents an initial estimate of geometric parameters representing each primitive type through the voting process, localizes the search of the optimal solution in a dimensionality-reduced parameter space, and increases the number of primitives that the method can discover. However, it is also limited by the category of geometric primitives. Parametric space clustering methods include [13], which uses integral images to compute local surface normals, cluster, segment, and classify points in normal space and spherical coordinates. It has good robustness to outliers but is restricted to low dimensions. Clustering-based methods are mainly divided into two categories, namely, region growing-based methods and automatic clustering methods. Region growing-based methods usually start from the seed point [14], [15] and extend the region through similarity constraints such as Euclidean distance [16], [17], [18], normal direction [14], [19], and curvature [20], [21] to form primitives. Rabbani et al. [22]

Received 22 September 2025; accepted 10 October 2025. Date of publication 13 November 2025; date of current version 16 December 2025. This work was supported by Sichuan Science and Technology Program under Grant 2025YFHZ0034. The Associate Editor coordinating the review process was Dr. Damodar Reddy Edla. (Corresponding author: Junpeng Xue.)

Ran Jia, Kelei Wang, Jiashi Zeng, and Yusong Meng are with the School of Aeronautics and Astronautics, Sichuan University, Chengdu, Sichuan 610207, China.

Junpeng Xue is with the School of Aeronautics and Astronautics, Sichuan University, Chengdu, Sichuan 610207, China, and also with the Robotic Satellite Key Laboratory of Sichuan Province, Sichuan University, Chengdu, Sichuan 610207, China (e-mail: jpxue@scu.edu.cn).

Feifei He is with Sichuan Aerospace Changzheng Equipment Manufacturing Company Ltd., Chengdu, Sichuan 610100, China.

Hongyang Chen is with Xichang Satellite Launch Center, Xichang, Sichuan 615000, China.

Digital Object Identifier 10.1109/TIM.2025.3632457

1557-9662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: UNIVERSIDAD DE LEON. Downloaded on March 22, 2026 at 09:57:32 UTC from IEEE Xplore. Restrictions apply.

is a region growing algorithm based on curvature smoothness constraint, but it is sensitive to seed selection, noise, and outliers. The method based on automatic clustering gathers points with similar characteristics into the same cluster, which is suitable for irregular point clouds. Xie et al. [23] preliminarily divide the point cloud through local quadric fitting and DBSCAN algorithm and then solve the oversegmentation problem in Mb-DBSCAN [24] through merging processing. However, the normal vector adjustment step does not take into account the influence of noise points.

To sum up, the current geometry-based point cloud classification methods surface three problems: 1) limited by the types of geometric primitives; 2) sensitive to noise; and 3) need to adjust more parameters for different types of objects.

B. Learning-Based Segmentation Methods

With the development of deep learning, many methods for point cloud segmentation have emerged. While point cloud segmentation methods for radar scenes [25], [26], [27] and semantic segmentation of point clouds [28] at conventional resolutions have been extensively studied, research on point cloud instance segmentation algorithms remains scarce, particularly for high-resolution scenarios. The pioneer of learning-based instance segmentation is ParseNet [29], which segments the input point cloud into primitive-type clusters by a decomposition module, followed by a fitting module that estimates the parameters of each instance. HPNet [30] uses the dense descriptor and obtains consistency feature and smoothness feature through the spectral embedding module, and finally realizes instance segmentation through mean-shift clustering. However, spectral calculation involves matrix computing, which places higher requirements on hardware devices when the point cloud resolution is high. PrimitiveNet [31] introduces the idea of determining whether two adjacent points are in the same instance for the first time, and realizes segmentation by region growing. It only discriminates the local part of the point cloud, so it is not limited by the number of input points. However, its input requires the triangulation surface reconstruction of the point cloud in advance, which increases the complexity of engineering applications. SEDNet [32] proposes a dual-branch network that extracts type, edge, and geometric features, detects primitives and edge points through feature fusion, and reconstructs the patches through a joint fitting strategy. PBPS [33] proposes a point cloud segmentation network aided by point cloud boundary probability and improves primitive extraction by probabilistically modeling boundary ambiguity using Gaussian distributions and adaptive feature processing. However, both SEDNet and PBPS are strictly limited in the number of input point clouds and cannot directly process high-resolution point clouds but need to be downsampled. This results in the loss of local detail structure of the original point cloud.

In addition to the aforementioned work that treats instance segmentation as a fundamental 3-D vision task, some methods have designed efficient segmentation algorithms tailored for specific scenarios. CAIS [34] proposed a center-aware instance segmentation method for close objects in industrial 3-D point cloud scenes. 3DGraphSeg [35] designed a unified graph

representation-based semantic segmentation network for high-speed railway environments. Some methods design point cloud segmentation algorithms for pipelines [36] and cantilever structure [37]; other point cloud segmentation methods designed for different application scenarios include [38], [39], and [40], but none of them are suitable for high-resolution point cloud instance segmentation. Notably, the paradigm of unsupervised learning has demonstrated significant potential in addressing the data dependency challenge across various vision tasks. In 2-D computer vision, for instance, numerous unsupervised and self-supervised methods have been developed to effectively utilize unlabeled data and reduce annotation requirements [41], [42], [43]. Similarly, in the context of railway infrastructure inspection, multimodal and self-supervised learning methods have shown significant progress in addressing challenges such as limited annotated data and complex environmental conditions. For example, DDPM-MILNet [44] proposes a denoising diffusion probabilistic model (DDPM) pretrained on unlabeled images combined with audio signals to achieve robust pantograph–catenary arc detection with minimal labeled samples. CSRM-MIM [45] introduces a self-supervised pretraining framework using a semantic masking strategy and a Siamese network to effectively utilize unlabeled catenary support component images. Furthermore, in surrogate modeling of pantograph–catenary system interactions, PhyFnoNet [46] is proposed to achieve accurate response prediction and state identification without relying heavily on large labeled datasets. These approaches demonstrate how semi-supervised, self-supervised learning, and advanced learning paradigms can enhance model generalization and reduce annotation dependency in specialized industrial applications, yet they remain constrained by domain-specific architectures and are not directly applicable to high-resolution, category-agnostic point cloud instance segmentation.

Learning-based segmentation methods also face three problems, in addition to the limited type of primitives that exist with geometry-based segmentation methods: 1) input point cloud resolution is limited and 2) dataset training and expensive hardware equipment are required.

To address the challenges mentioned above, we propose a novel ridge-patterned instance segmentation method for high-resolution point clouds. As illustrated in Fig. 1, our method demonstrates superior performance and strong noise robustness compared to existing approaches, achieving high segmentation accuracy in both noise-free and noisy environments. Our contributions can be summarized as follows.

- 1) We develop a robust ridge-based segmentation framework that scales effectively to point clouds of varying sizes and resolutions while maintaining parameter stability across different scales and complex geometries.
- 2) We present a curvature coefficient estimation and ridge extraction technique using the determinantal sampling (DetS) estimator, which demonstrates remarkable resilience to noise while preserving geometric fidelity.
- 3) We propose an adaptive ridge enhancement and surface clustering pipeline that combines ridge points denoising with an adaptive-radius region growing algorithm to achieve precise and contiguous segmentation of surface point clouds.

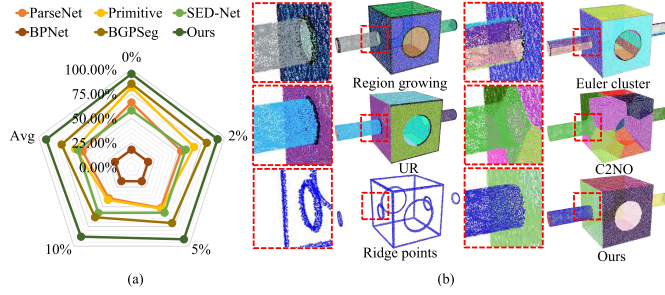


Fig. 1. Comparison between our method and other methods. Our approach achieves excellent segmentation results both in noise-free and noisy conditions as shown in the radar graph, demonstrating superior robustness and accuracy. (a) mIoU for noise experiment. (b) Comparison of segmentation results.

- 4) We introduce a local plane fitting strategy for ridge point assignment that significantly improves segmentation accuracy by mitigating boundary artifacts and misclassification through geometric consistency optimization.

The rest of this article is organized as follows. Section II presents the detailed methodology of our ridge-patterned instance segmentation approach, including the robust ridge point extraction using the DetS estimator, ridge point enhancement, adaptive-radius region growing algorithm, and local plane fitting strategy for ridge assignment. Section III describes the experimental setup and results, including performance comparisons with both geometry- and learning-based methods on synthetic (ABC-10) and real-world (OmniObject3D-8) datasets, noise experiments, ablation studies validating each key component of our method, and real-world workpiece experiments. Finally, Section IV concludes this article with a summary of findings and a discussion of potential future research directions.

II. METHOD

A. Overview

Our ridge-patterned instance segmentation algorithm processes high-resolution point clouds through a four-step framework as illustrated in Fig. 2, designed to overcome key challenges in noise robustness and parameter stability. The process begins by robustly extracting ridge points using a DetS estimator, which leverages curvature coefficients calculated via a noise-resistant triangular threshold method, addressing a key limitation of existing approaches in noisy environments. Next, we enhance ridge points' connectivity through a density-aware denoising step and virtual point interpolation, ensuring topological continuity even for fragmented boundaries. To partition the remaining surface points, we introduce an adaptive-radius region growing algorithm that dynamically adjusts the search radius based on proximity to ridge points, enabling precise clustering without manual parameter tuning. Finally, ridge points are assigned to instances via local plane fitting, a strategy that mitigates boundary artifacts by minimizing geometric deviation rather than relying on nearest neighbor points. This unified pipeline achieves resolution-independent segmentation while maintaining stability across diverse noise levels and object scales.

B. Robust Ridge Point Extraction Based on DetS Estimator

To accomplish point cloud instance segmentation, the proposed method initially separates the point cloud into ridge points and discrete surface points. The process begins by computing a robust noise-resistant curvature coefficient using both the DetS estimator and principal component analysis (PCA). Subsequently, a triangular threshold method is employed to segment the point cloud into distinct ridge points and discrete surface points.

Given the high sensitivity of curvature coefficients to noise, we first compute a robust optimal covariance matrix for each point in the point cloud using a DetS estimator as shown in Fig. 3. For each point $\mathbf{x}_i \in \mathcal{P}$, we randomly sample m subsets from its k_1 -nearest neighbors

$$\mathcal{S}_j^{(i)} = \{\mathbf{x}_{j_1}, \mathbf{x}_{j_2}, \dots, \mathbf{x}_{j_s}\} \subset \mathcal{P}, \quad j = 1, \dots, m \quad (1)$$

where $|\mathcal{S}_j^{(i)}| = s$ and $\mathbf{x}_i \in \mathcal{S}_j^{(i)}$. $\mathcal{S}_j^{(i)}$ represent the subset of k_1 -nearest neighbors points. For each $\mathcal{S}_j^{(i)}$, we compute the mean vector $\boldsymbol{\mu}_j$ and covariance matrix $\boldsymbol{\Sigma}_j$ as follows:

$$\boldsymbol{\mu}_j = \frac{1}{s} \sum_{l=1}^s \mathbf{x}_{j_l} \quad \boldsymbol{\Sigma}_j = \frac{1}{s-1} \sum_{l=1}^s (\mathbf{x}_{j_l} - \boldsymbol{\mu}_j)(\mathbf{x}_{j_l} - \boldsymbol{\mu}_j)^\top. \quad (2)$$

The Mahalanobis distance is then computed for each point. Using the 97.5% quantile of the chi-square distribution as the threshold $\chi_{0.975}^2(3) \approx 9.348$, points with distances below this threshold are identified as inner points

$$D_M^2(\mathbf{x}_{j_l}) = (\mathbf{x}_{j_l} - \boldsymbol{\mu}_j)^\top \boldsymbol{\Sigma}_j^{-1} (\mathbf{x}_{j_l} - \boldsymbol{\mu}_j) \quad (3)$$

$$\mathcal{I}_j = \left\{ \mathbf{x}_{j_l} \in \mathcal{S}_j^{(i)} \mid D_M^2(\mathbf{x}_{j_l}) < \chi_{0.975}^2(3) \right\} \quad (4)$$

where $\mathcal{I}_i$ represents the set of inner points. If the determinant of the current subset's covariance matrix is smaller than the current minimum determinant and the subset contains more points than the required minimum, then the corresponding covariance matrix for that point is updated.

After obtaining a robust covariance matrix for each point, we can calculate the eigenvalues of the current point on three axes as $\lambda_1 > \lambda_2 > \lambda_3$. The curvature coefficient ω is defined as

$$\omega = \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3}. \quad (5)$$

Points with larger curvature coefficients exhibit sharper features and are identified as ridge points. We employ a triangular threshold method to extract the initial ridge points through the following procedure. First, construct a frequency distribution histogram of the point cloud's curvature coefficients. Second, define a straight line L connecting the histogram's highest peak to its minimum frequency point. Finally, compute the perpendicular distance d_i from each ordinate to the line L as

$$d_i = \frac{|Ax_i + Bh(x_i) + C|}{\sqrt{A^2 + B^2}} \quad \forall i \in \{1, \dots, N\}. \quad (6)$$

The curvature threshold ω_{th} is determined by selecting the abscissa corresponding to the maximum perpendicular distance in the histogram. Points with curvature coefficients satisfying $x_i > \omega_{th}$ are classified as ridge points

$$\mathcal{P}_{ridge}^{ori} = \{\mathbf{p}_{x_i} \mid x_i > \omega_{th}\}, \quad \text{where } \omega_{th} = \arg \max_{x_i} d_i \quad (7)$$

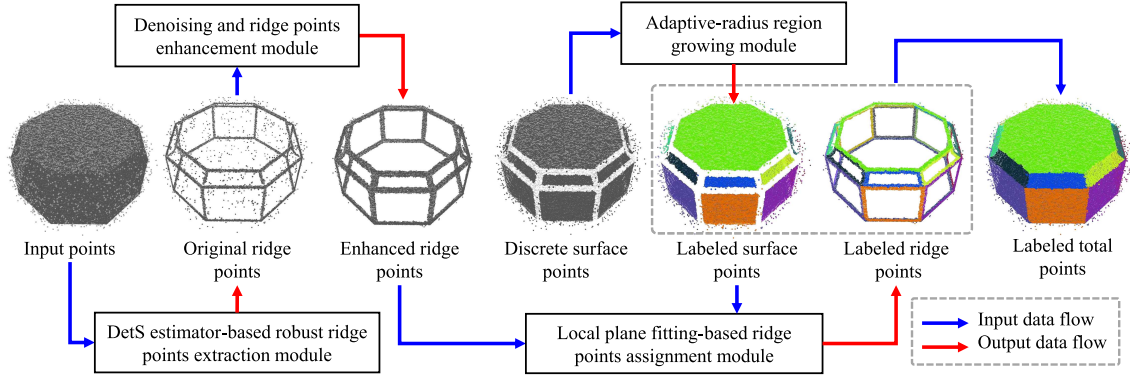


Fig. 2. Pipeline for robust ridge model point cloud instance segmentation, consisting of four key steps: ridge point extraction using the DetS estimator, ridge point enhancement, adaptive-radius region growing segmentation, and ridge point assignment via local plane fitting.

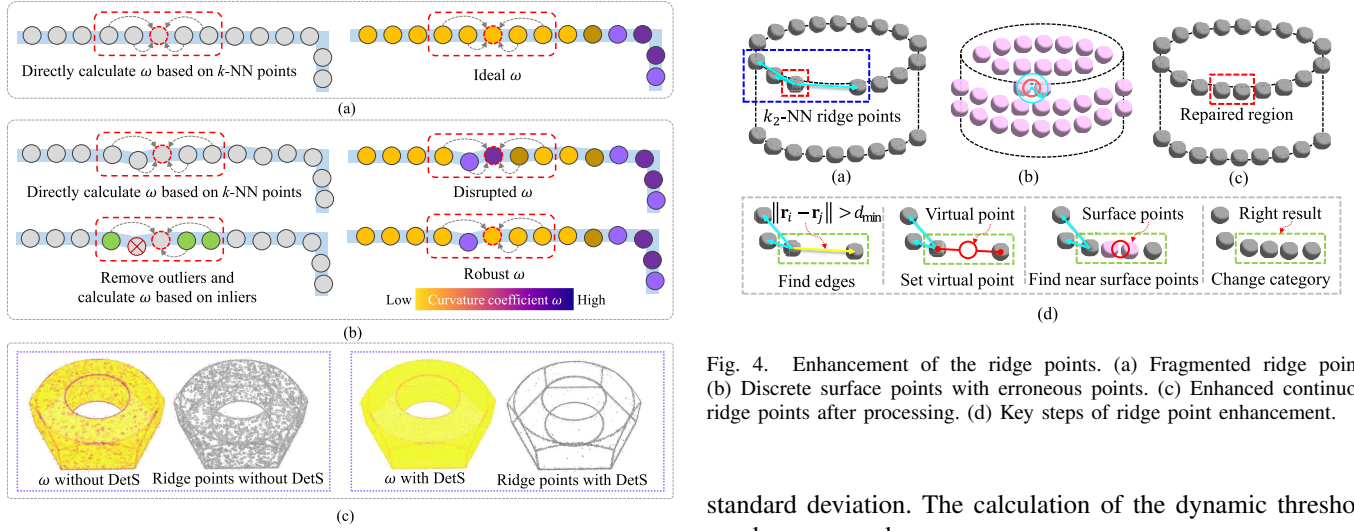


Fig. 3. DetS estimator's role in curvature coefficient computation. (a) Curvature coefficient calculation process under ideal conditions. (b) Perturbation of curvature coefficients under noisy conditions while highlighting the advantage of robust curvature coefficients computed using inliers selected by the DetS estimator. (c) Distribution differences of curvature coefficients with/without the DetS estimator under noise. Yellow indicates lower curvature coefficients, while purple represents higher values.

where $\mathcal{P}_{\text{ridge}}^{\text{ori}}$ represents the set of original ridge points.

C. Ridge Points Denoising and Enhancement

Since the ridge extraction algorithm based on the DetS estimator filters out both noise points and ridge points, $\mathcal{P}_{\text{ridge}}^{\text{ori}}$ contains not only ridge points but also noise points. Therefore, we need to further process $\mathcal{P}_{\text{ridge}}^{\text{ori}}$, that is, remove outliers according to the density of the point cloud. First, calculate the average distance of each point's k_1 -nearest neighbors and normalize it

$$\bar{d}_i = \frac{\bar{d}_i - \min_m \bar{d}_m}{\max_n \bar{d}_n - \min_m \bar{d}_m}, \text{ where } \bar{d}_i = \frac{1}{k_1 - 1} \sum_{j=2}^{k_1} d(\mathbf{p}_i, \mathbf{p}_{i,j}). \quad (8)$$

Calculate the density weight for each point. The smaller the average distance, the larger the corresponding weight, and vice versa. In addition, the global threshold needs to be calculated and dynamically adjusted according to the weights. The global threshold is obtained by jointly calculating the mean and

Fig. 4. Enhancement of the ridge points. (a) Fragmented ridge points. (b) Discrete surface points with erroneous points. (c) Enhanced continuous ridge points after processing. (d) Key steps of ridge point enhancement.

standard deviation. The calculation of the dynamic threshold can be expressed as

$$T_i = T_{\text{global}} (\kappa + \tau (1 - w_i)), \text{ where } T_{\text{global}} = \mu + \lambda \sigma \\ w_i = e^{-\bar{d}_i} \quad (9)$$

where κ and τ represent the weight coefficient, μ is the mean value of the average distance, σ is the standard deviation, and λ is the $n-\sigma$ parameter. The point where the average distance is less than the threshold is the final ridge point

$$\mathcal{P}_{\text{ridge}}^{\text{filt}} = \{\mathbf{p}_i \in \mathcal{P}_{\text{ridge}}^{\text{ori}} | \bar{d}_i < T_i\}. \quad (10)$$

At this stage, the point cloud has been partitioned into three distinct sets: the ridge point set, the discrete surface point set, and the noise point set. For ease of representation, these three sets are represented as $\mathcal{R} = \{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N\}$, $\mathcal{F} = \{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_M\}$, and $\mathcal{S} = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_H\}$.

The subsequent region growing algorithm utilizes Euclidean distance as the connectivity criterion for point clouds. However, the ridge points extracted through the aforementioned procedures frequently present discontinuity artifacts. Therefore, enhancement of the initially extracted ridge points becomes necessary to enhance the methodological robustness.

The ridge point enhancement procedure, illustrated in Fig. 4, consists of the following steps. For each ridge point $\mathbf{r}_i \in \mathcal{R}$, we first identify its k_2 -nearest neighbors within the ridge

point set and construct connecting edges between $\mathbf{r}_i$ and each neighboring point

$$\mathcal{E} = \{(\mathbf{r}_i, \mathbf{r}_j) \mid \mathbf{r}_j \in \mathcal{N}_{k_2}(\mathbf{r}_i) \ \forall \mathbf{r}_i \in \mathcal{R}\}, \text{ where } \mathcal{N}_{k_2}(\mathbf{r}_i) \subseteq \mathcal{R}. \quad (11)$$

For each constructed edge, we calculate its length and select edges whose lengths exceed the shortest edge's threshold

$$\mathcal{E}_{\text{valid}} = \{(\mathbf{r}_i, \mathbf{r}_j) \in \mathcal{E} \mid \|\mathbf{r}_i - \mathbf{r}_j\| > d_{\min}\}. \quad (12)$$

The threshold calculation process for the shortest edge distance $d_{\min}$ proceeds as follows. First, we compute the distances from each discrete surface point $\mathbf{f}_i$ to its k_3 -nearest neighbors, constructing a frequency distribution histogram $H(\mathcal{D}, \text{bins} = B)$ of the resulting distance set $\mathcal{D}$. The threshold $d_{\min}$ is then determined as the distance value corresponding to the highest peak in $H(\mathcal{D}, \text{bins} = B)$

$$d_{\min} = \arg \max_d H(d, \text{bins} = B). \quad (13)$$

For each valid edge $(\mathbf{r}_i, \mathbf{r}_j) \in \mathcal{E}_{\text{valid}}$ satisfying the criteria, virtual ridge points are sampled at predetermined intervals along the edge, with the number of sampled points being proportional to the edge length

$$\mathcal{V}_{ij} = \left\{ \mathbf{r}_i + t \cdot \frac{\mathbf{r}_j - \mathbf{r}_i}{L_{ij}} \mid t = \delta, 2\delta, \dots, n_{ij}\delta \right\} \quad (14)$$

where $n_{ij} = \lfloor L_{ij}/\delta \rfloor$ and $L_{ij} = \|\mathbf{r}_j - \mathbf{r}_i\|$. δ is the sampling interval. The virtual point set is defined as $\mathcal{V} = \bigcup_{(\mathbf{r}_i, \mathbf{r}_j) \in \mathcal{E}_{\text{valid}}} \mathcal{V}_{ij}$. For each virtual ridge point $\mathbf{v} \in \mathcal{V}$, we then identify its k_{dyn} -nearest neighbors within the discrete surface point set $\mathcal{F}$, where the parameter k_{dyn} is proportional to the corresponding edge length

$$\mathcal{N}_{k_{\text{dyn}}}(\mathbf{v}) \subseteq \mathcal{F}, \text{ where } k_{\text{dyn}} = \lfloor \alpha \cdot L_{ij} \rfloor \quad (15)$$

where α is the proportional coefficient. Finally, add the searched discrete surface points to the ridge point set

$$\mathcal{R}_{\text{en}} = \mathcal{R} \cup \mathcal{F}_{\text{add}}, \text{ where } \mathcal{F}_{\text{add}} = \bigcup_{\mathbf{v} \in \mathcal{V}} \mathcal{N}_{k_{\text{dyn}}}(\mathbf{v}). \quad (16)$$

The enhanced ridge points are more continuous and have a larger range, which makes the fracture between the point clouds of the discrete surface more significant and facilitates the subsequent segmentation steps of the region growing algorithm. For ease of representation, these three sets after the above enhancement are represented as $\mathcal{R}_{\text{en}}$, $\mathcal{F}_{\text{en}}$, and $\mathcal{S}$.

D. Clustering Discrete Surface Points Through Adaptive-Radius Region Growing Method

After extracting the enhanced ridge points, we obtain unlabeled discrete surface point blocks, which are then classified using our adaptive-radius region growing algorithm. Calculate a unique search radius for each point before performing radius-based region growing segmentation on the discrete surface point clouds. Categories containing relatively few points are reclassified into the ridge point set to complete the process.

The segmentation steps of the discrete surface points are shown in Algorithm 1. In the first step, we need to calculate the search radius of each point $\mathbf{f}_i \in \mathcal{F}_{\text{en}}$. Find the k_3 -nearest neighbors of each point $\mathbf{f}_i$, count the distance from the point to

Algorithm 1 Discrete Surface Point Cloud Classification and Ridge Points Updating

Input: $\mathcal{F}_{\text{en}}$: Discrete surface points set, $\mathcal{R}_{\text{en}}$: Ridge points set, C : Curvature coefficients

Output: $\mathcal{L}$: Class labels for discrete surface points, $\mathcal{R}_{\text{new}}$: Updated ridge points set

// Step 1: Compute search radius for each point

Compute KNN distances $D = \bigcup_{i=1}^n \{\text{distance}(\mathbf{f}_i, \mathbf{f}_j) \mid \mathbf{f}_j \in \text{KNN}(\mathbf{f}_i, k_3)\}$

Compute histogram $H \leftarrow \text{Histogram}(D, \text{bins} = B)$

Initialize search radius $r_{\text{init}} \leftarrow \arg \max_d H(d)$

foreach $\mathbf{f}_i \in \mathcal{F}_{\text{en}}$ **do**

$d_{\min}(\mathbf{f}_i) \leftarrow \min_{\mathbf{r} \in \mathcal{R}} \|\mathbf{f}_i - \mathbf{r}\|$

$x_i \leftarrow \text{Normalize}(d_{\min}(\mathbf{f}_i))$ // Normalize to $[0, 1]$

$w_i \leftarrow a + b \cdot x_i$ // Compute weight

$r_i \leftarrow r_{\text{init}} \cdot w_i$ // Weighted radius

// Step 2: Classify surface points

$\mathcal{L} \leftarrow \emptyset$ // Initialize class labels

$U \leftarrow \mathcal{F}_{\text{en}}$ // Unclassified points

while $U \neq \emptyset$ **do**

$\mathbf{f}^* \leftarrow \arg \min_{\mathbf{f}_i \in U} c_i$ // Select point with minimum curvature

$L \leftarrow \{\mathbf{f}^*\}$ // Initialize current class

$T \leftarrow [\mathbf{f}^*]$ // Processing queue

while $T \neq \emptyset$ **do**

$\mathbf{f}_j \leftarrow T.\text{pop}()$

$N_j \leftarrow \{\mathbf{f}_k \in U \mid \|\mathbf{f}_j - \mathbf{f}_k\| \leq r_j\}$ // Find neighbors

$L \leftarrow L \cup N_j$

$T \leftarrow T \cup N_j$

$U \leftarrow U \setminus N_j$ // Remove from the unclassified set

$\mathcal{L} \leftarrow \mathcal{L} \cup \{L\}$ // Add to classified set

// Step 3: Filter small classes and update ridge points

$\mathcal{R}_{\text{new}} \leftarrow \mathcal{R}_{\text{en}}$

foreach $L \in \mathcal{L}$ **do**

if $|L| < M$ **then**

$\mathcal{R}_{\text{new}} \leftarrow \mathcal{R}_{\text{new}} \cup L$ // Merge small classes into ridge

return $\mathcal{L}, \mathcal{R}_{\text{new}}$

the k_3 -nearest neighbors, and construct a frequency distribution histogram of the distance. The distance corresponding to the histogram with the highest frequency is set to the initialization radius r_{init} . The calculation steps of the initialization radius are similar to the calculation of $d_{\min}$ in Section II-C. Since our region growing algorithm relies only on the Euclidean distance to judge the connectivity, we need to treat the search radius more strictly.

An apparently valid assumption is that the search radius of the point can be larger at a far distance from the ridge point, but the search should be performed more cautiously at a near distance from the ridge point, that is, the search radius of the point should be smaller. Therefore, we define the weight of the search radius by linear weighting. In the second step, we implement the segmentation of the discrete surface points

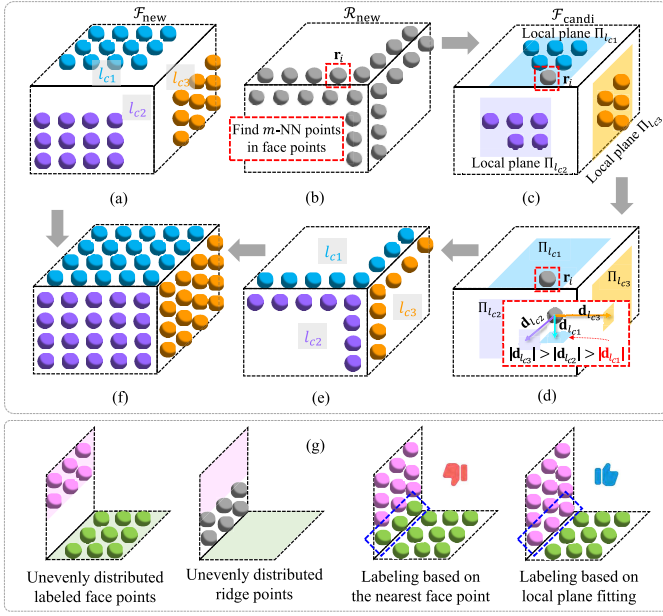


Fig. 5. Assignment of the ridge points through local plane fitting. (a)–(f) Process of assignment. (g) Special case demonstrating the necessity and superiority of the local plane fitting.

through a region growing algorithm based on the weighted adaptive search radius. In addition, we choose the point with the smallest coefficient of curvature in the cloud of undivided points as the starting search point. In the third step, we treat point clouds with too few points as misclassifications and add them to the ridge point set. In addition, the set of noise points $\mathcal{S}$ is classified as ridge points. After this step, updated ridge points $\mathcal{R}_{new}$, surface points $\mathcal{F}_{new}$ and surface points' labels $\mathcal{L}$ can be obtained.

E. Assigning the Ridge Points Through Local Plane Fitting

After the discrete surface points are segmented, the extracted ridge points need to be classified. In order to resist the aliasing phenomenon caused by the classification of the nearest neighbor surface point cloud, we design a classification method of ridge points based on local plane fitting. For each ridge point $\mathbf{r}_i \in \mathcal{R}_{new}$, find m -nearest neighbors point in $\mathcal{F}_{new}$, and count the unique set of labels

$$\mathcal{F}_{candi} = \{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_m\} \subset \mathcal{F}_{new} \quad (17)$$

$$\mathcal{L}_{candi} = \text{Unique}\{L(\mathbf{f}_1), L(\mathbf{f}_2), \dots, L(\mathbf{f}_m)\} \subset \mathcal{L} \quad (18)$$

where Unique() is the deduplicate operation, $\mathcal{F}_{candi}$ is the m -nearest neighbors point set of $\mathbf{r}_i$, corresponding to the blue, purple, and orange balls in Fig. 5(a). For each candidate label $l_c \in \mathcal{L}_{candi}$, select the points in the surface point set whose label is l_c

$$\mathcal{F}_{LPF}^j = \{\mathbf{f}_{candi} \in \mathcal{F}_{new} | L(\mathbf{f}_{candi}) = l_c\}, \quad j = 0, 1, \dots, |\mathcal{L}_{candi}| \quad (19)$$

where $|\mathcal{L}_{candi}|$ is the number of labels in $\mathcal{L}_{candi}$. Find p -nearest neighbors in $\mathcal{F}_{LPF}^j$ for $\mathbf{r}_i$, which are used to fit local plane.

TABLE I
DETAILS OF THE ABC-10 DATASET

Index	CAD	Faces	Points	X-scale	Y-scale	Z-scale
1		15	97,992	5.46	5.06	1.59
2		15	65,970	21.34	9.53	9.53
3		22	148,890	4.29	4.18	4.17
4		18	101,358	57.15	57.15	6.35
5		18	112,350	120.90	120.90	55.76
6		13	132,810	32.00	20.00	44.00
7		32	133,944	20.57	6.99	26.29
8		11	84,570	25.40	63.50	25.40
9		12	170,004	50.00	60.00	60.00
10		15	98,550	127.00	40.31	38.10

The plane parameters are obtained through the least-squares method

$$\Pi_{l_c} : \min_{\|\mathbf{n}\|=1, d} \sum_{m=1}^p (\mathbf{n} \cdot \mathbf{f}_m + d)^2, \quad \text{where } \mathbf{f}_m \in \mathcal{F}_{LPF-candi}^j \quad (20)$$

where $\mathcal{F}_{LPF-candi}^j$ represents the set of p -nearest neighbors in $\mathcal{F}_{LPF}^j$. $\mathbf{n}$ is the unit normal vector, d is the vertical intercept, and Π_{l_c} is the fit local plane with the label of l_c . The distance from $\mathbf{r}_i$ to the local plane Π_{l_c} can be defined as

$$|\mathbf{d}_{l_c}| = D(\mathbf{r}_i, \Pi_{l_c}) = |\mathbf{n} \cdot \mathbf{r}_i + d| \quad (21)$$

where $\mathbf{d}_{l_c}$ is the normal vector from $\mathbf{r}_i$ to local plane Π_{l_c} . Finally, choose the best label l_c^* , which corresponds to the minimum distance $D(\mathbf{r}_i, \Pi_{l_c})$ as the final label of $\mathbf{r}_i$.

Compared with directly finding the label of the nearest surface point as the label of the $\mathbf{r}_i$, the method based on local plane fitting can more faithfully select the correct label. We show the superiority of our method in special cases in Fig. 5(g). In the case of unsatisfactory distribution of surface points and ridge points, our method has the advantage of resisting jagged noise.

III. EXPERIMENTS

A. Dataset

To test the effectiveness of our method, we tested it on synthetic data and real scan data. To test the segmentation effect on the standard artifact point cloud, we constructed a high-resolution subset ABC-10 on the basis of the ABC synthetic dataset [47], which contains ten point clouds with representative shape structures at different scales, and their

details are shown in Table I. To simulate the point cloud situation in real-world scanning, we constructed a high-resolution subset OmniObject3D-8 on the basis of the OmniObject3D [48] dataset, which contains eight actual point clouds with measurement errors. Since the ABC-10 dataset is an enhanced version generated by upsampling the ABC dataset, while the OmniObject3D-8 dataset is directly selected from the original OmniObject3D collection, this article only elaborates on the construction details of ABC-10. For comprehensive information about the OmniObject3D-8 dataset, please refer to the original publication [48].

B. Metrics

In order to comprehensively evaluate the performance of the segmentation algorithm, for each class C_i , we use precision, recall, $F1$ -score, and mean intersection over Union as evaluation metrics

$$P = \frac{1}{N_{\text{cla}}} \sum_{i=1}^{N_{\text{cla}}} p_i, \quad p_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \quad (22)$$

$$R = \frac{1}{N_{\text{cla}}} \sum_{i=1}^{N_{\text{cla}}} r_i, \quad r_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \quad (23)$$

$$F1 = \frac{1}{N_{\text{cla}}} \sum_{i=1}^{N_{\text{cla}}} f_i, \quad f_i = 2 \times \frac{P_i \times R_i}{P_i + R_i} \quad (24)$$

$$\text{mIoU} = \frac{1}{N_{\text{cla}}} \sum_{i=1}^{N_{\text{cla}}} \text{IoU}_i, \quad \text{IoU}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \quad (25)$$

where N_{cla} is the number of instance classes; true positive TP means that the sample belongs to the C_i class and is correctly predicted as the C_i class; false positive FP means that the sample does not belong to the C_i class but is incorrectly predicted as the C_i class; and false negative FN means that the sample belongs to the C_i class, but is incorrectly predicted as other categories.

In addition, we also calculated AP_{25} , AP_{50} , and AP in ablation experiments and noise robustness experiments, which can cover the evaluation of point cloud segmentation algorithms under different stringencies. AP_{25} and AP_{50} denote the average precision scores evaluated at IoU thresholds of 25% and 50%, respectively. The metric AP is defined as the mean of these scores across a range of IoU thresholds from 50% to 95%, incremented in steps of 5% [49], [50].

C. Implementation Details

All experiments are conducted on a workstation equipped with a 13th Gen Intel Core i9-13900K CPU and 64 GB of RAM. While an NVIDIA GeForce RTX 4090 GPU was used for fair and efficient benchmarking of learning-based comparative methods, we emphasize that our proposed method is entirely unsupervised and operates efficiently on CPU-only environments, making it accessible for deployment on resource-constrained systems.

The parameter settings for our method are as follows: the neighborhood size $k_1 = 30$, $k_2 = 20$, and $k_3 = 5$, the bin count $B = 100$ in triangular threshold method, the sampling interval $\delta = 0.1$, the proportional coefficient $\alpha = 10$, the

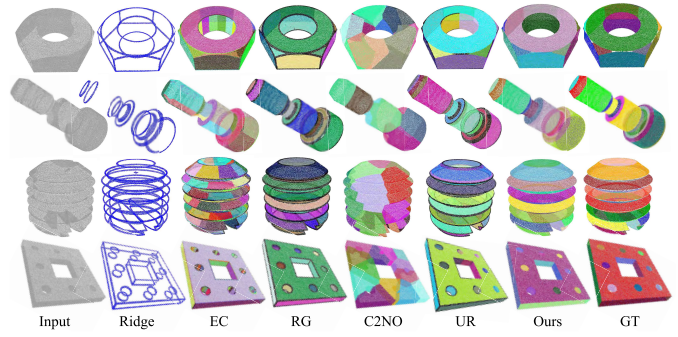


Fig. 6. Visualization results compared with geometry-based methods.

enhancement weights $\kappa = 0.9$ and $\tau = 0.1$ for ridge points, and the coefficients $a = 1.6$ and $b = 0.05$ in Algorithm 1.

D. Results on ABC-10 Dataset

1) *Comparison With Geometry-Based Methods:* We evaluate four geometry-based point cloud segmentation algorithms—RG [22], EC [51], C2NO [52], and UR [53]—on the ABC-10 dataset. The input data consist exclusively of raw 3-D coordinates, with all normal vectors required by these algorithms being computed via PCA.

The segmentation results are visualized in Fig. 6, revealing distinct limitations of each baseline method. EC demonstrates susceptibility to cylindrical structure detection failures attributable to its fixed angle threshold, while our method dynamically adapts to capture complete cylindrical and arc features. RG exhibits boundary segmentation deficiencies stemming from normal vector computation inaccuracies, while our approach eliminates the need for precise normal vectors. C2NO underperforms due to its uniform segmentation criterion being incompatible with the inherent category size variations in point clouds, while our approach automatically adjusts to inherent point cloud variations. UR can quickly achieve good segmentation results, but it suffers from missed detections in boundary regions. Moreover, for irregular surfaces such as the conical frustum side in object 9, it may produce incorrect segmentation.

Using the evaluation metrics defined in Section III-B (with results presented in Table II), we observe that EC and RG demonstrate higher precision on partial point clouds, achieved through their inherent tendency to minimize FPs. However, this precision comes at the expense of increased FNs.

A method can achieve high precision by adopting an overly conservative strategy (i.e., reducing FPs at the cost of increasing FNs). This leads to undersegmentation, where significant portions of valid surfaces are missed, resulting in low recall. This phenomenon is consistent with the results in Table II, where EC and RG achieved SOTA precision but accompanied by low recall.

In contrast, both $F1$ -score and mIoU penalize both FPs and FNs. A high mIoU or $F1$ -score necessarily requires both precision and recall to be high simultaneously, which is a more challenging and desirable goal. Our method simultaneously maintains both low FP and FN rates, achieving superior performance across all mIoU and $F1$ -score metrics, demonstrating its comprehensive segmentation quality.

TABLE II
QUANTITATIVE RESULTS COMPARED WITH GEOMETRY-BASED METHODS

No.	Method	P	R	F1	mIoU	No.	Method	P	R	F1	mIoU
1	EC	78.49%	72.44%	72.44%	64.87%	6	EC	90.80%	58.20%	58.20%	49.86%
	RG	100.00%	71.91%	82.21%	71.91%		RG	68.37%	59.68%	62.50%	57.18%
	C2NO	45.39%	28.57%	30.43%	18.39%		C2NO	33.10%	25.43%	24.06%	14.16%
	UR	98.24%	88.17%	91.79%	86.66%		UR	98.18%	88.53%	92.74%	86.83%
	Ours	99.56%	99.15%	99.35%	98.71%		Ours	95.42%	99.37%	96.94%	94.81%
2	EC	51.13%	41.87%	41.87%	28.76%	7	EC	74.28%	47.44%	47.44%	38.79%
	RG	81.62%	63.48%	65.36%	58.89%		RG	67.68%	59.52%	58.98%	55.92%
	C2NO	49.55%	37.30%	37.20%	24.50%		C2NO	63.38%	25.36%	33.96%	20.90%
	UR	78.08%	75.56%	74.82%	69.06%		UR	97.74%	88.67%	92.68%	86.72%
	Ours	82.53%	99.39%	89.00%	81.97%		Ours	93.80%	99.34%	94.47%	93.18%
3	EC	99.99%	32.99%	32.99%	32.99%	8	EC	99.95%	72.28%	72.28%	72.23%
	RG	99.99%	73.04%	83.48%	73.03%		RG	100.00%	83.95%	90.87%	83.95%
	C2NO	27.71%	23.83%	24.90%	14.85%		C2NO	39.45%	40.10%	38.97%	28.28%
	UR	100.00%	77.56%	86.65%	77.56%		UR	100.00%	89.91%	94.02%	89.91%
	Ours	96.32%	99.68%	97.85%	96.03%		Ours	98.72%	99.69%	99.19%	98.42%
4	EC	100.00%	55.24%	55.24%	55.24%	9	EC	99.88%	51.86%	51.86%	51.77%
	RG	100.00%	78.41%	87.84%	78.41%		RG	100.00%	85.36%	91.96%	85.36%
	C2NO	56.57%	20.75%	24.97%	14.45%		C2NO	27.26%	29.78%	22.42%	13.18%
	UR	100.00%	81.09%	88.28%	81.09%		UR	99.99%	57.40%	69.51%	57.39%
	Ours	99.88%	98.04%	98.94%	97.92%		Ours	98.90%	99.62%	99.25%	98.51%
5	EC	91.96%	82.27%	82.27%	76.62%	10	EC	98.97%	77.89%	77.89%	77.11%
	RG	99.99%	86.52%	92.69%	86.51%		RG	100.00%	83.68%	90.96%	83.68%
	C2NO	59.09%	42.19%	42.25%	28.48%		C2NO	43.23%	36.03%	35.18%	21.84%
	UR	99.88%	88.87%	93.26%	88.77%		UR	99.99%	89.46%	93.91%	89.46%
	Ours	99.12%	97.75%	98.42%	96.92%		Ours	99.63%	99.08%	99.35%	98.71%

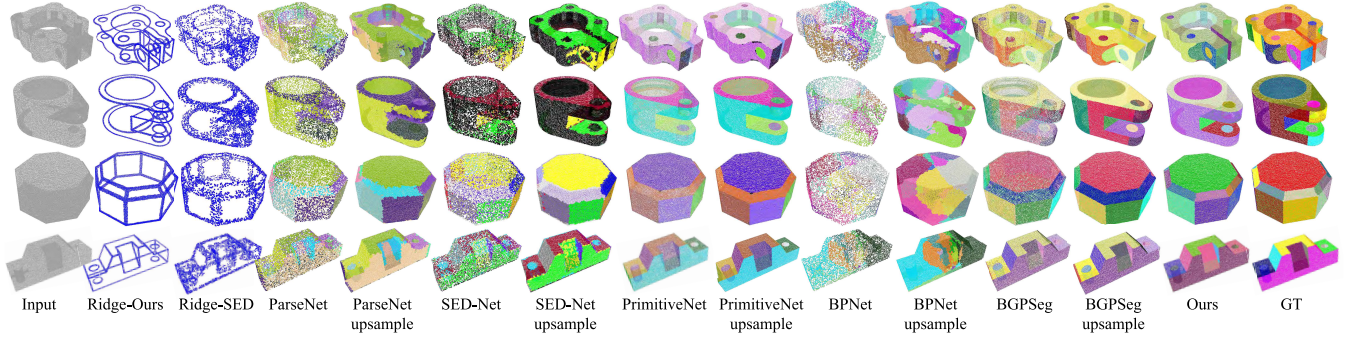


Fig. 7. Visualization results compared with learning-based methods, including both low-resolution point clouds and upsampled point clouds.

Furthermore, these comparative methods exhibit pronounced parameter sensitivity, with their parameter optimization representing an NP-hard problem that significantly limits practical applicability. In contrast, our method maintains consistent parameters setting across the entire ABC-10 dataset, effectively handling point clouds with diverse scales and geometric characteristics without requiring parameter adjustments. This parameter invariance demonstrates our approach's strong robustness against geometric variations.

2) *Comparison With Learning-Based Methods:* To benchmark our method against deep learning approaches, we evaluate five state-of-the-art point cloud instance segmentation algorithms: ParseNet [29], PrimitiveNet [31], SED-Net [32], BGPSeg [54], and BPNet [55]. PrimitiveNet and BGPSeg's input format includes point cloud connectivity information, so we utilized its native data directly. However, SED-Net (trained on 10 000 input points), ParseNet (trained on 10 000 input points), and BPNet (trained on 8192 input points) require downsampling of the ABC-10 dataset. Notably, we also evaluated SED-Net's boundary detection capability for

comparative analysis. To ensure fair evaluation of the learning-based methods' metrics, we upsample the segmentation results based on the nearest neighbor principle as in [31].

Fig. 7 shows partial visualization results, including segmentation outputs from both low-resolution point clouds and upsampled point clouds. The results demonstrate consistent undersegmentation across all five learning-based methods. A typical example is the incorrect lid segmentation observed on octagonal boxes. In contrast, our method achieves correct segmentation. Furthermore, the boundary computation results of SED-Net show significantly poorer performance compared to our ridge point extraction approach. As shown in the first two rows of Fig. 7, both other learning-based methods and our approach exhibit limitations in regions where curvature changes are gradual or indistinct, with the exception of BGPSeg. This represents a critical direction for future improvements in both geometry- and learning-based segmentation methods.

The quantitative results of learning-based methods are presented in Table III. Our method achieves optimal performance

TABLE III
QUANTITATIVE RESULTS COMPARED WITH LEARNING-BASED METHODS

No.	Method	P	R	F1	mIoU	No.	Method	P	R	F1	mIoU
1	ParseNet	90.96%	97.40%	93.99%	88.78%	6	ParseNet	69.96%	98.01%	78.84%	68.45%
	Primitive	97.63%	97.67%	97.64%	95.45%		Primitive	67.30%	69.16%	65.78%	63.37%
	SED-Net	76.27%	82.38%	73.40%	67.16%		SED-Net	64.03%	87.27%	71.88%	56.96%
	BPNet	58.29%	63.82%	55.90%	46.24%		BPNet	30.54%	27.07%	23.17%	13.82%
	BGPSeg	96.81%	89.46%	92.61%	86.74%		BGPSeg	95.29%	95.97%	95.55%	91.52%
	Ours	99.56%	99.15%	99.35%	98.71%		Ours	95.42%	99.37%	96.94%	94.81%
2	ParseNet	71.89%	98.07%	81.11%	70.51%	7	ParseNet	66.62%	94.06%	74.98%	63.00%
	Primitive	81.15%	82.66%	80.19%	75.62%		Primitive	91.27%	92.22%	86.55%	85.33%
	SED-Net	64.06%	83.55%	69.61%	57.62%		SED-Net	56.03%	88.21%	61.07%	50.24%
	BPNet	30.34%	25.29%	25.50%	16.71%		BPNet	23.53%	27.44%	20.54%	13.03%
	BGPSeg	76.04%	78.73%	76.20%	69.38%		BGPSeg	83.96%	85.60%	84.37%	79.79%
	Ours	82.53%	99.39%	89.00%	81.97%		Ours	93.80%	99.34%	94.47%	93.18%
3	ParseNet	24.03%	43.89%	26.53%	15.58%	8	ParseNet	92.69%	96.41%	94.14%	89.37%
	Primitive	59.90%	53.38%	46.46%	42.03%		Primitive	98.89%	99.42%	99.15%	98.33%
	SED-Net	72.38%	74.73%	61.53%	48.84%		SED-Net	88.23%	87.97%	85.74%	76.70%
	BPNet	33.96%	27.02%	28.40%	20.03%		BPNet	48.27%	31.89%	31.86%	19.90%
	BGPSeg	94.95%	96.29%	95.54%	91.56%		BGPSeg	97.79%	94.96%	96.22%	92.84%
	Ours	96.32%	99.68%	97.85%	96.03%		Ours	98.72%	99.69%	99.19%	98.42%
4	ParseNet	78.18%	95.39%	84.63%	75.03%	9	ParseNet	39.65%	97.89%	52.36%	38.52%
	Primitive	99.02%	98.91%	98.96%	97.95%		Primitive	99.21%	99.21%	99.21%	98.43%
	SED-Net	54.95%	88.53%	64.78%	52.24%		SED-Net	48.44%	82.68%	60.85%	43.97%
	BPNet	28.15%	35.19%	24.55%	15.64%		BPNet	29.85%	20.33%	19.99%	11.23%
	BGPSeg	98.75%	90.98%	94.63%	89.83%		BGPSeg	96.42%	94.10%	95.07%	90.79%
	Ours	99.88%	98.04%	98.94%	97.92%		Ours	98.90%	99.62%	99.25%	98.51%
5	ParseNet	85.18%	93.58%	88.73%	81.02%	10	ParseNet	76.42%	93.63%	81.99%	71.97%
	Primitive	72.67%	88.50%	73.51%	70.28%		Primitive	90.94%	92.54%	91.62%	90.19%
	SED-Net	80.52%	92.77%	81.43%	73.67%		SED-Net	61.11%	85.19%	67.92%	53.50%
	BPNet	31.09%	31.19%	23.06%	14.29%		BPNet	17.51%	23.95%	15.26%	9.18%
	BGPSeg	84.53%	85.05%	80.09%	75.62%		BGPSeg	93.80%	91.91%	92.64%	86.41%
	Ours	99.12%	97.75%	98.42%	96.92%		Ours	99.63%	99.08%	99.35%	98.71%

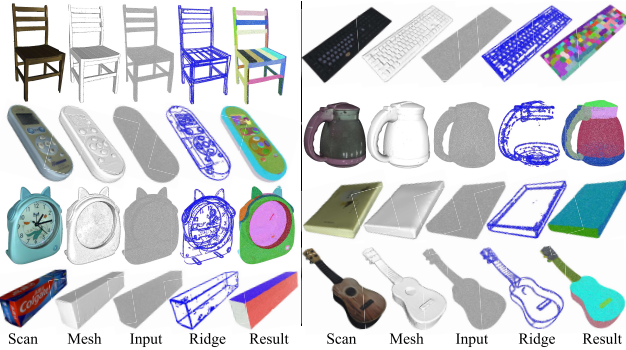


Fig. 8. Visualization results on OmniObject3D-8 dataset.

across multiple metrics for the majority of objects. It should be noted that, compared to learning-based approaches, our method possesses an inherent advantage in handling point clouds with arbitrary resolution and unconstrained inputs. This capability is particularly crucial for segmenting high-resolution point clouds of precision components.

E. Results on OmniObject3D-8 Dataset

To validate our method's instance segmentation performance on real-world scanned point clouds, we selected eight samples from the OmniObject3D dataset for testing, as shown in Fig. 8. Each panel displays the original scanned mesh, the mesh without RGB information, the input point cloud, our method's extracted ridge points, and the final instance segmentation results.

Real-world scanned meshes typically contain significant measurement noise and complex geometries that substantially increase segmentation difficulty. Our method demonstrates

notable robustness to such noise through two key mechanisms. While noise points may occasionally be captured during ridge point extraction, they are effectively resolved in the subsequent assignment phase without compromising global segmentation quality. For structurally complex objects like keyboards and remote controls, the method maintains precise segmentation capability, successfully distinguishing individual keys and buttons despite their fine-scale features.

F. Noise Experiments

To evaluate the performance of our method under noisy conditions, we add artificial random noise to the ABC-10 dataset and test it. We add noise from two dimensions: the number of noise points and the offset. We choose the number of noise points at 2%, 5%, and 10%, and the offset scale of noise points at 0.01, 0.02, and 0.03. The offset of the noise point is calculated by the oriented bounding box (OBB) of the input point cloud. Given an input point cloud with OBB edge lengths $a_1 > a_2 > a_3$ and a specified offset scale β , the absolute noise displacement is calculated as

$$\delta = a_1 \times \beta. \quad (26)$$

We compared its results against those of other methods at different noise levels. We evaluated four geometry-based methods and five learning-based methods under noisy conditions, with noise levels set at 2%, 5%, and 10% under an offset scale of $\beta = 0.02$. It is worth noting that methods such as EC, RG, and UR are capable of detecting noise points, which is a valuable capability in practical applications. However, in our experimental setup, this ability leads to a decrease in their quantitative metrics. Since the recognition

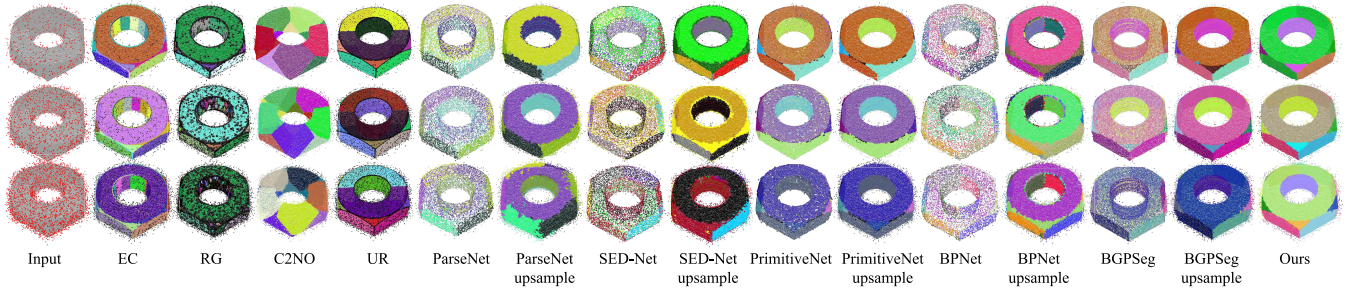


Fig. 9. Visualization results of noise comparison experiments, including both low-resolution point clouds and upsampled point clouds.

TABLE IV
QUANTITATIVE RESULTS OF NOISE COMPARISON EXPERIMENTS

Method	2%-0.02				5%-0.02				10%-0.02			
	P	R	F1	mIoU	P	R	F1	mIoU	P	R	F1	mIoU
C2NO	44.19%	29.44%	30.17%	18.80%	45.41%	30.03%	30.88%	19.31%	43.98%	29.61%	30.27%	19.00%
ParseNet	59.18%	85.03%	65.69%	53.51%	56.86%	85.04%	64.45%	51.54%	45.07%	75.44%	52.55%	39.34%
Primitive	73.83%	76.41%	71.03%	67.33%	66.07%	64.27%	57.11%	52.55%	61.01%	54.08%	46.08%	41.14%
SED-Net	67.07%	86.48%	70.75%	58.34%	66.41%	86.02%	70.21%	57.53%	66.65%	85.38%	70.53%	57.73%
BPNet	33.01%	31.22%	26.74%	17.93%	32.65%	31.14%	26.65%	17.85%	32.86%	30.81%	26.37%	17.61%
BGPSeg	87.70%	86.16%	86.23%	80.92%	79.23%	78.45%	76.48%	70.61%	79.40%	71.65%	69.78%	63.35%
Ours	94.40%	96.19%	94.82%	92.56%	94.12%	95.95%	94.38%	91.18%	92.82%	93.48%	92.29%	87.81%

of noise points penalizes their segmentation scores under our evaluation protocol, we opted to exclude the quantitative results of these three methods from the performance table. Instead, we provide visual comparisons to objectively reflect their segmentation behavior. As shown in Fig. 9, we compare the segmentation results of our method against others on a hex screw workpiece across different noise levels. Table IV further reports quantitative metrics of each method under noise. It should be noted that the reported P , R , $F1$, and $mIoU$ values represent the average precision, average recall, average $F1$ -score, and average mean intersection over union, respectively, across all object segmentation results for each method. Both quantitative and visualization results demonstrate that our method achieves superior segmentation performance in noisy environments.

To more intuitively demonstrate the performance degradation of various methods under increasing noise levels, we visualize the changes in evaluation metrics using clustered bar charts, as shown in Fig. 10. Within each cluster, the rate of change in the bars corresponding to different noise levels reflects how rapidly each method degrades with increasing noise. Experimental results show that our method exhibits significantly slower and more limited performance degradation compared to other approaches, demonstrating strong robustness even under high-noise conditions.

Furthermore, we examined the performance of our approach under more finely graded noise intensities and across varying evaluation strictness criteria. The quantitative results of the noise robustness experiment are shown in Table V. As the noise ratio increases, $mIoU$, AP_{25} , AP_{50} , and AP decreases. However, within the reasonable noise range commonly encountered in practical applications, all metrics remain at a high level, demonstrating strong practical robustness. We visualize the noise experimental results of two point clouds in the ABC-10 dataset as shown in Fig. 11, including the noise contained in the input point cloud, the undenoised and

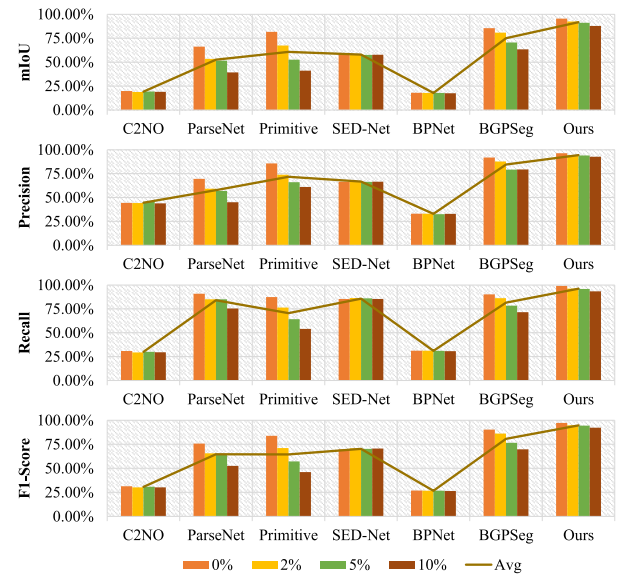


Fig. 10. Statistics results of noise comparison experiments.

enhanced ridge point extraction results, and the final example segmentation results. The experimental results demonstrate that our method maintains accurate ridge point extraction across all tested noise conditions, with segmentation performance showing minimal degradation even at maximum noise levels (10% point contamination with $\beta = 0.03$ scale offset).

G. Ablation Experiments

In order to validate the effectiveness of our proposed robust ridge point extraction module, ridge point enhancement module, and ridge points cloud assignment module based on local plane fitting, we conducted multiple ablation experiments in different noise environments.

The ablation experiments we designed are tested for four noise cases, including no noise point, 2% noise points, 5%

TABLE V
QUANTITATIVE RESULTS OF NOISE ROBUSTNESS ANALYSIS

Percent	Scale	mIoU	AP ₂₅	AP ₅₀	AP
2%	0.01	92.55%	84.21%	81.87%	81.29%
	0.02	92.56%	83.63%	82.46%	81.51%
	0.03	92.59%	83.63%	82.46%	81.58%
2%-Average		92.57%	83.82%	82.26%	81.46%
5%	0.01	90.66%	82.46%	81.29%	79.24%
	0.02	91.18%	83.04%	81.29%	79.39%
	0.03	90.58%	83.63%	82.46%	81.07%
5%-Average		90.81%	83.04%	81.68%	79.90%
10%	0.01	89.74%	81.86%	78.36%	76.39%
	0.02	87.81%	83.03%	80.70%	78.87%
	0.03	83.77%	81.87%	80.70%	77.78%
10%-Average		87.11%	82.25%	79.92%	77.68%
w/o noise		95.52%	85.96%	84.21%	82.60%

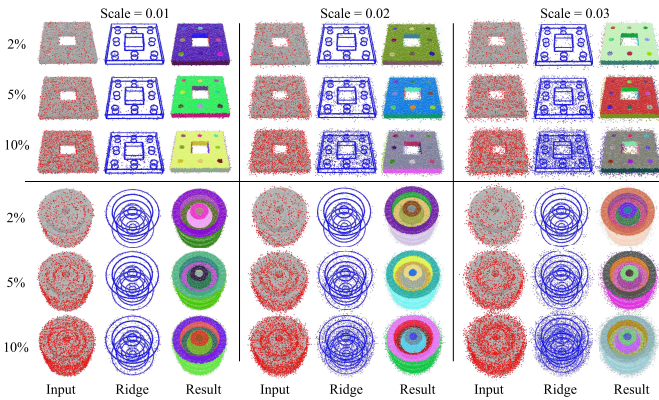


Fig. 11. Visualization results of noise robustness analysis.

TABLE VI
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS

Percent	Model	DetS	Fit	Enh	mIoU	AP ₂₅	AP ₅₀	AP
0%	A	✓			94.07%	82.46%	80.70%	78.80%
	B	✓	✓		95.02%	82.46%	80.70%	78.87%
	C	✓	✓	✓	95.52%	85.96%	84.21%	82.60%
2%	A	✓			92.03%	78.36%	76.02%	75.15%
	B	✓	✓		93.25%	78.36%	76.02%	75.15%
	C	✓	✓	✓	92.56%	83.63%	82.46%	81.51%
5%	A	✓			90.92%	76.02%	74.85%	73.54%
	B	✓	✓		92.06%	78.36%	76.02%	75.15%
	C	✓	✓	✓	91.18%	83.04%	81.29%	79.39%
10%	A	✓			87.82%	76.02%	73.10%	71.71%
	B	✓	✓		88.82%	76.02%	73.10%	72.00%
	C	✓	✓	✓	87.81%	83.03%	80.70%	78.87%

noise points, and 10% noise points. The offset scale of the noise points is 0.02, and the definition of the offset scale can be referred to (26). The ablation experiments consist of three models. Model A only uses the robust ridge extraction module (DetS). Model B adds a ridge point assignment module (Fitting, abbreviated as “Fit”) based on local plane fitting on the basis of Model A, and Model C adds the ridge point enhancement module (Enhance, abbreviated as “Enh”) on the basis of Model B. The quantitative results are shown in Table VI. Since the DetS module is very important, the lack of it will lead to segmentation failure, so we did not design the experiment without the DetS module. It can be found that compared with Model A, Model B can achieve the improvement of mIoU under different noise conditions because the ridge point assignment strategy based on local

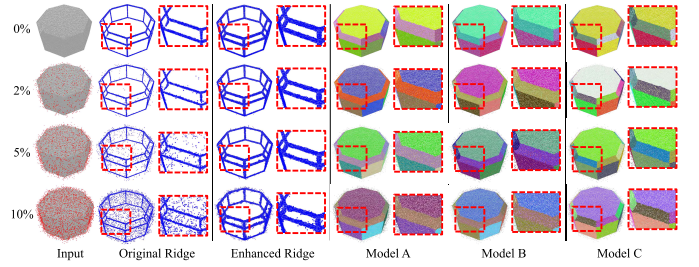


Fig. 12. Visualization results of ablation experiments.

plane fitting can faithfully select the correct category, thereby improving the boundary division accuracy. Compared with Model B, Model C's ridge point enhancement module can robustly ensure the independence of point cloud block partition, significantly improving AP₂₅, AP₅₀, and AP.

At the same time, Model C may experience a decrease in mIoU, which is due to the increase in the number of ridge points due to ridge point enhancement, which increases the difficulty of ridge point assignment. The addition of the Enhance module sacrifices the slight decrease in mIoU, resulting in a significant improvement in AP metrics. In addition, we visualize a representative point cloud in the ABC-10 dataset in Fig. 12, showing the ridge point extraction effect and instance segmentation effect of different models in detail.

It can be seen that under different noise conditions, the closed ridge points cannot be extracted from the lid part of the octagonal box, and the ridge points through denoising and enhancement are strongly connected, which can accurately divide the point cloud blocks. Although in the case of high-noise ratio, the enhanced ridge points are also accompanied by noise, this phenomenon does not affect the subsequent distribution steps and has little impact on the segmentation results.

H. Real-World Workpiece Experiments

To further verify the segmentation effectiveness of our method on real-world workpiece point clouds, we performed measurements and tests on a standard Flange workpiece. The point cloud data of the Flange is captured using a dual camera-projector system. The measurement environment and equipment are illustrated in Fig. 13(a). The system includes two CMOS cameras with a resolution of 1600×1200 , model UI-3250CP-M-GL from IDS Imaging Development Systems GmbH in Obersulm, Germany, along with a DLP projector capable of 30 frames/s and a resolution of 1280×800 , model PDC03-A manufactured by Giant Vinda in Fuzhou, China. Furthermore, as the Flange workpiece is metallic in nature, as shown in Fig. 13(b), we used scanning spray to coat the metal surface, producing a spray-coated Flange depicted in Fig. 13(c), in order to enhance its diffuse reflectance.

We measured the top surface of the Flange and acquired a high-resolution point cloud consisting of 514469 points, as shown in Fig. 13(d). The segmentation results and the extracted ridge points are presented in Fig. 13(e) and (f), respectively. It can be noted that the ridge points exhibit discontinuities. This occurs because the fractured regions are

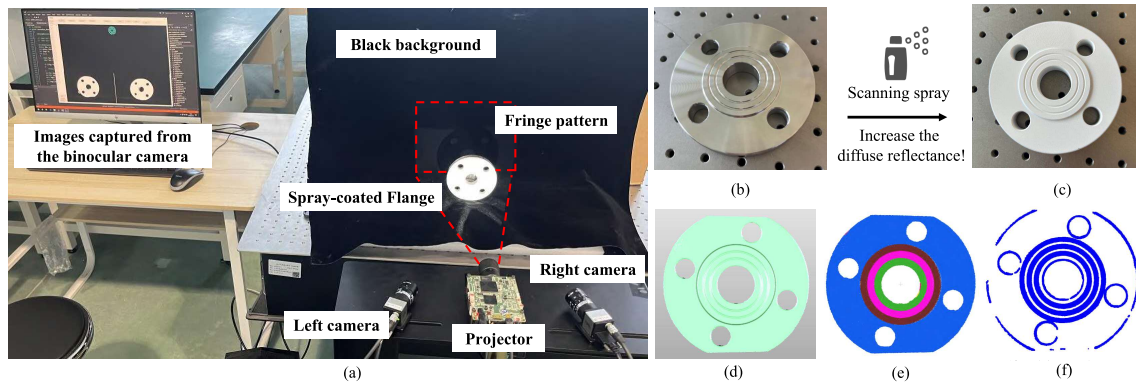


Fig. 13. Segmentation results for real-world workpiece point cloud using a dual camera-projector system. (a) Dual camera-projector system. (b) Metal flange. (c) Spray-coated flange. (d) Point cloud. (e) Result. (f) Ridge points.

situated on the outer edges or the inner walls of the holes. When measuring from a single angle, only portions of the point cloud blocks with significant curvature variations could be captured.

IV. CONCLUSION

Our work presents a novel ridge-patterned instance segmentation method for high-resolution point cloud, addressing key limitations in existing approaches, such as noise sensitivity and parameter instability. By integrating curvature-based ridge extraction, ridge enhancement, adaptive-radius region growing, and ridge assignment, the proposed algorithm achieves robust segmentation without relying on primitive fitting or supervised training. The experimental results validate the method's effectiveness across diverse datasets, including synthetic and real-world point clouds. Notably, it maintains strong performance under significant noise, demonstrating its practical applicability in challenging environments. The algorithm's parameter stability and resolution independence further enhance its usability in industrial applications like precision measurement and reverse engineering.

While the proposed method demonstrates robust performance in various scenarios, it still exhibits certain limitations under specific challenging conditions. The segmentation performance may be suboptimal on surfaces with extremely smooth and gradual curvature variations due to the lack of distinct ridge features, as discussed in Section III-D. Furthermore, in the presence of substantial noise, the curvature estimation step requires significantly more iterations, which hinders real-time performance in demanding environments. Additionally, the method remains susceptible to densely distributed noise that can interfere with the initial ridge point extraction process. For example, as shown in Fig. 8, the ridge extraction of the toothpaste box was notably affected by localized dense noise. Moreover, the current adaptive mechanism shows limited capability in handling complex thin structures, particularly under constrained point cloud resolution. These limitations highlight potential directions for future methodological improvements.

Future improvements will focus on three key directions: developing adaptive curvature thresholding techniques to better handle objects with smooth surfaces, investigating more

advanced robust statistical methods to enhance performance in high-noise environments, and exploring multiscale region growing strategies to improve segmentation of complex structures with varying densities and scales. These refinements are expected to extend the method's applicability across more challenging real-world scenarios.

ACKNOWLEDGMENT

The authors would like to thank Zi Guo from the Nuclear Power Institute of China for the helpful discussions.

REFERENCES

- [1] B. Chen et al., "Hyperspectral LiDAR point cloud segmentation based on geometric and spectral information," *Opt. Exp.*, vol. 27, no. 17, pp. 24043–24059, 2019.
- [2] P. Tian and X. Hua, "Robust surface area measurement of unorganized point clouds based on multiscale supervoxel segmentation," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–17, 2024.
- [3] A. Kaiser, J. A. Y. Zepeda, and T. Boubekeur, "A survey of simple geometric primitives detection methods for captured 3D data," *Comput. Graph. Forum*, vol. 38, no. 1, 2018, pp. 167–196.
- [4] R. Schnabel, R. Wahl, and R. Klein, "Efficient RANSAC for point-cloud shape detection," in *Computer Graphics Forum*. Hoboken, NJ, USA: Wiley, 2007, pp. 214–226.
- [5] Y. Li, X. Wu, Y. Chrysathou, A. Sharf, D. Cohen-Or, and N. J. Mitra, "GlobFit: Consistently fitting primitives by discovering global relations," in *Proc. ACM SIGGRAPH Papers*, Jul. 2011, pp. 1–12.
- [6] Z. Kang and Z. Li, "Primitive fitting based on the efficient multiBaySAC algorithm," *PLoS ONE*, vol. 10, no. 3, Mar. 2015, Art. no. e0117341.
- [7] P. V. Hough, "Method and means for recognizing complex patterns," U.S. Patent 3,069,654, Washington, DC, USA, Dec. 18, 1962.
- [8] R. Hulik, M. Spanel, P. Smrz, and Z. Materna, "Continuous plane detection in point-cloud data based on 3D Hough transform," *J. Vis. Commun. Image Represent.*, vol. 25, no. 1, pp. 86–97, Jan. 2014.
- [9] T. Rabbani and F. V. D. Heuvel, "Efficient Hough transform for automatic detection of cylinders in point clouds," *ISPRS J. Photogramm. Remote Sens.*, vol. 3, pp. 60–65, Jan. 2005.
- [10] F. A. Limberger and M. M. Oliveira, "Real-time detection of planar regions in unorganized point clouds," *Pattern Recognit.*, vol. 48, no. 6, pp. 2043–2053, Jun. 2015.
- [11] L. A. F. Fernandes and M. M. Oliveira, "Real-time line detection through an improved Hough transform voting scheme," *Pattern Recognit.*, vol. 41, no. 1, pp. 299–314, Jan. 2008.
- [12] A. Raffo, S. Romanengo, B. Falcidieno, and S. Biasotti, "Fitting and recognition of geometric primitives in segmented 3D point clouds using a localized voting procedure," *Comput. Aided Geometric Design*, vol. 97, Jan. 2022, Art. no. 102123.
- [13] D. Holz, S. Holzer, R. B. Rusu, and S. Behnke, "Real-time plane segmentation using rgb-d cameras," in *Robot Soccer World Cup*. Cham, Switzerland: Springer, 2011, pp. 306–317.

- [14] S. Oesau, F. Lafarge, and P. Alliez, "Planar shape detection and regularization in tandem," in *Computer Graphics Forum*. Hoboken, NJ, USA: Wiley, 2015, pp. 203–215.
- [15] Y. Zhou, K. Yin, H. Huang, H. Zhang, M. Gong, and D. Cohen-Or, "Generalized cylinder decomposition," *ACM Trans. Graph.*, vol. 34, no. 6, pp. 1–171, Nov. 2015.
- [16] R. F. Salas-Moreno, B. Glocken, P. H. J. Kelly, and A. J. Davison, "Dense planar SLAM," in *Proc. IEEE Int. Symp. Mixed Augmented Reality (ISMAR)*, Sep. 2014, pp. 157–164.
- [17] T. Shao, W. Xu, K. Zhou, J. Wang, D. Li, and B. Guo, "An interactive approach to semantic modeling of indoor scenes with an RGBD camera," *ACM Trans. Graph.*, vol. 31, no. 6, pp. 1–11, Nov. 2012.
- [18] H. Lin et al., "Semantic decomposition and reconstruction of residential scenes from LiDAR data," *ACM Trans. Graph.*, vol. 32, no. 4, pp. 1–10, Jul. 2013.
- [19] A. Monszpart, N. Mellado, G. J. Brostow, and N. J. Mitra, "Rapter: Rebuilding man-made scenes with regular arrangements of planes," *ACM Trans. Graph. (TOG)*, vol. 34, no. 4, pp. 1–103, 2015.
- [20] Y. Zhang, W. Xu, Y. Tong, and K. Zhou, "Online structure analysis for real-time indoor scene reconstruction," *ACM Trans. Graph.*, vol. 34, no. 5, pp. 1–13, Nov. 2015.
- [21] O. Mattausch, D. Panozzo, C. Mura, O. Sorkine-Hornung, and R. Pajarola, "Object detection and classification from large-scale cluttered indoor scans," *Comput. Graph. Forum*, vol. 33, no. 2, pp. 11–21, May 2014.
- [22] T. Rabbani, F. Van Den Heuvel, and G. Vosselmann, "Segmentation of point clouds using smoothness constraint," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 36, no. 5, pp. 248–253, 2006.
- [23] T. Xie, H. Chen, W. Liu, R. Zhou, and Q. Li, "3D surface segmentation from point clouds via quadric fits based on DBSCAN clustering," *Pattern Recognit.*, vol. 154, Oct. 2024, Art. no. 110589.
- [24] H. Chen, M. Liang, W. Liu, W. Wang, and P. X. Liu, "An approach to boundary detection for 3D point clouds based on DBSCAN clustering," *Pattern Recognit.*, vol. 124, Apr. 2022, Art. no. 108431.
- [25] F. Wang, Z. Wu, Y. Yang, W. Li, Y. Liu, and Y. Zhuang, "Real-time semantic segmentation of LiDAR point clouds on edge devices for unmanned systems," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.
- [26] Y. An, W. Liu, Y. Cui, J. Wang, X. Li, and H. Hu, "Multilevel ground segmentation for 3-D point clouds of outdoor scenes based on shape analysis," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022.
- [27] X. Du, Y. Lu, and Q. Chen, "A fast multiplane segmentation algorithm for sparse 3-D LiDAR point clouds by line segment grouping," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–15, 2023.
- [28] Y. Cui, Y. An, W. Sun, H. Hu, and X. Song, "Lightweight attention module for deep learning on classification and segmentation of 3-D point clouds," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021.
- [29] G. Sharma, D. Liu, S. Maji, E. Kalogerakis, S. Chaudhuri, and R. Měch, "ParSeNet: A parametric surface fitting network for 3D point clouds," in *Proc. ECCV*, Aug. 2020, pp. 261–276.
- [30] S. Yan, Z. Yang, C. Ma, H. Huang, E. Vouga, and Q. Huang, "HPNet: Deep primitive segmentation using hybrid representations," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2733–2742.
- [31] J. Huang, Y. Zhang, and M. Sun, "PrimitiveNet: Primitive instance segmentation with local primitive embedding under adversarial metric," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15323–15333.
- [32] Y. Li, S. Liu, X. Yang, J. Guo, J. Guo, and Y. Guo, "Surface and edge detection for primitive fitting of point clouds," in *Proc. Special Interest Group Comput. Graph. Interact. Techn. Conf. Conf.*, Jul. 2023, pp. 1–10.
- [33] S. Wang, F. Qin, Y. Tong, X. Shang, and Z. Zhang, "Probabilistic boundary-guided point cloud primitive segmentation network," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.
- [34] J. Zhang, Z. Zhou, and J. Sun, "Center-aware instance segmentation for close objects in industrial 3-D point cloud scenes," *IEEE Trans. Ind. Informat.*, vol. 20, no. 2, pp. 2812–2821, Feb. 2024.
- [35] Y. Geng et al., "3DGraphSeg: A unified graph representation-based point cloud segmentation framework for full-range high-speed railway environments," *IEEE Trans. Ind. Informat.*, vol. 19, no. 12, pp. 11430–11443, Dec. 2023.
- [36] J. Zhang, Z. Zhou, and J. Sun, "Three-dimensional segmentation and global clearance analysis for free-bent pipelines in point-cloud scenarios," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.
- [37] Z. Han, C. Yang, and Z. Liu, "Cantilever structure segmentation and parameters detection based on concavity and convexity of 3-D point clouds," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 6, pp. 3026–3036, Jun. 2020.
- [38] Z. Wang et al., "Self-attentive local aggregation learning with prototype guided regularization for point cloud semantic segmentation of high-speed railways," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 11157–11170, Oct. 2023.
- [39] Y. Li, Y. Wang, and Y. Liu, "Three-dimensional point cloud segmentation based on context feature for sheet metal part boundary recognition," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–10, 2023.
- [40] J. Wang, C. Xu, L. Dai, J. Zhang, and R. Zhong, "An unequal deep learning approach for 3-D point cloud segmentation," *IEEE Trans. Ind. Informat.*, vol. 17, no. 12, pp. 7913–7922, Dec. 2021.
- [41] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.
- [42] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.
- [43] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.
- [44] J. Yan et al., "Research on multimodal techniques for arc detection in railway systems with limited data," *Structural Health Monitor.*, May 2025, doi: 10.1177/14759217251336797.
- [45] H. Yang et al., "CSRM-MIM: A self-supervised pretraining method for detecting catenary support components in electrified railways," *IEEE Trans. Transport. Electrification*, vol. 11, no. 4, pp. 10025–10037, Aug. 2025.
- [46] Y. Cheng et al., "Surrogate modeling of pantograph-catenary system interactions," *Mech. Syst. Signal Process.*, vol. 224, Feb. 2025, Art. no. 112134.
- [47] S. Koch et al., "ABC: A big CAD model dataset for geometric deep learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9593–9603.
- [48] T. Wu et al., "OmniObject3D: Large-vocabulary 3D object dataset for realistic perception, reconstruction and generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 803–814.
- [49] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.
- [50] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
- [51] R. B. Rusu and S. Cousins, "3D is here: Point cloud library (PCL)," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2011, pp. 1–4.
- [52] A. F. R. Guarda, N. M. M. Rodrigues, and F. Pereira, "Constant size point cloud clustering: A compact, non-overlapping solution," *IEEE Trans. Multimedia*, vol. 23, pp. 77–91, 2021.
- [53] C. Romanengo, D. Cabiddu, S. Pittaluga, and M. Mortara, "Building semantic segmentation from large-scale point clouds via primitive recognition," *Graph. Models*, vol. 136, Dec. 2024, Art. no. 101234.
- [54] Z. Fang, C. Zhuang, Z. Lu, Y. Wang, L. Liu, and J. Xiao, "BGPSeg: Boundary-guided primitive instance segmentation of point clouds," *IEEE Trans. Image Process.*, vol. 34, pp. 1454–1468, 2025.
- [55] R. Fu, Q. Li, C. Wen, N. An, and F. Tang, "A novel framework for learning Bézier decomposition from 3D point clouds," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4329–4340, May 2025.

Ran Jia received the Bachelor of Engineering degree in spatial information and digital technology from China University of Geosciences, Wuhan, China, in 2023. She is currently pursuing the Master of Engineering degree in aerospace science and technology with Sichuan University, Chengdu, China. Her research interests include 3-D computer vision and vision measurement.

Junpeng Xue received the Ph.D. degree in optical engineering from Sichuan University, Chengdu, China, in 2012.

He is currently an Associate Professor at the School of Aeronautics and Astronautics, Sichuan University. In March 2016, he went to the Brookhaven National Laboratory, U.S. Department of Energy, for a one-year study visit. His main research interests include optical 3-D vision measurement and computer vision, involving high-precision 3-D shape measurement and detection of aerospace vehicles and aircraft pose measurement.

Dr. Xue is currently a member of the Optoelectronic Technology Special Committee of the Chinese Optical Society, the Director of Optoelectronic Technology and System Integration of China Instrument Society, and a member of Chinese Optical Society.

Feifei He received the bachelor's degree from Xihua University, Chengdu, China, in 2011, and the master's degree from Sichuan University, Chengdu, in 2014.

He is a Senior Engineer and works at Sichuan Aerospace Changzheng Equipment Manufacturing Company Ltd., Chengdu. His research interests include intelligent assembly.

Hongyang Chen received the master's degree in high-pressure science and technology from Sichuan University, Chengdu, China, in 2015.

He is a Senior Engineer and works at Xichang Satellite Launch Center. His research interests mainly focus on the research of technologies and applications in aerospace launch and intelligent assembly.

Kelei Wang received the Bachelor of Engineering degree in aircraft control and information engineering from Sichuan University, Chengdu, China, in 2024, where she is currently pursuing the Master of Engineering degree in aeronautical and astronautical science and technology.

Her research interests include vision measurement and computer vision.

Jiashi Zeng received the Bachelor of Engineering degree in mechanical design, manufacturing, and automation from Hunan Institute of Technology, Hengyang, China, in 2018. He is currently pursuing the Master of Engineering degree in aerospace engineering with Sichuan University, Chengdu, China.

His research interests include vision measurement and path planning.

Yusong Meng received the Bachelor of Software Engineering degree from Chengdu University of Technology, Chengdu, China, in 2025. He is currently pursuing the Master of Aeronautical Engineering with Sichuan University, Chengdu.

He is interested in computer vision and deep learning.